

### 3.3.1 Introduction to Differentiation

In sections 3.1 & 3.2 we found the derivatives of functions using *the definition of the derivative*. We noticed that this process was often very complicated and required a lot of work. Our goal in the following sections is to use our knowledge of *the definition of the derivative* to develop shortcuts in the process of finding the derivatives of various functions.

#### Recall:

**Definition:** The derivative of a function  $f$ , denoted by  $f'(x)$  is,

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

if this limit exists.

**Example 1:** In parts (a) through (c) find  $\frac{dy}{dx}$  using the definition of the derivative.

- a)  $y = x$
- b)  $y = x^2$
- c)  $y = x^3$
- d) Based upon your answers above, what would you guess is the first derivative of  $f(x) = x^4$ ? What about  $g(x) = x^{97}$ ?

*Solutions:*

a) We note that:  $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{x+h-x}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$

b) We note that:  $\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - x^2}{h} = \lim_{h \rightarrow 0} \frac{h(2x+h)}{h} = \lim_{h \rightarrow 0} (2x+h) = 2x$

c) We note that:

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{(x+h)^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h} \\ &= \lim_{h \rightarrow 0} (3x^2 + 3xh + h^2) = 3x^2 \end{aligned}$$

- d) It seems, in the parts above, that the power decreases by one and the old power becomes the coefficient. So, we would probably guess that  $f'(x) = 4x^3$  and  $g'(x) = 97x^{96}$

The next theorem will confirm that our guess in part (d) is correct, and it is of great importance.

**Theorem (The Power Rule):** If  $n$  is any real number, then  $\frac{d}{dx}(x^n) = nx^{n-1}$ .

*Note: There is a proof of this in the textbook when  $n$  is a natural number which uses the Binomial Theorem. We will need implicit differentiation, which we will learn about later, in order to give a complete proof.*

**Example 2:** Compute each of the following derivatives.

a)  $\frac{d}{dx}\left(\frac{1}{x}\right)$

We note:  $\frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = -1x^{-2} = \frac{-1}{x^2}$

b)  $\frac{d}{dt}(t^{1.7})$

We note:  $\frac{d}{dt}(t^{1.7}) = 1.7t^{0.7}$

c)  $\frac{d}{ds}(\sqrt[5]{s^7})$

We note:  $\frac{d}{ds}(\sqrt[5]{s^7}) = \frac{d}{ds}(s^{7/5}) = \frac{7}{5}s^{2/5}$

*Question: Could we use the power rule to figure out:  $\frac{d}{dx}(x^x)$ ?*

**Example 3:** Consider the curve given by  $f(x) = x\sqrt{x}$ .

a) Find the derivative of  $y$  with respect to  $x$ .

We note:  $f'(x) = \frac{d}{dx}(x\sqrt{x}) = \frac{d}{dx}(x^{3/2}) = \frac{3}{2}x^{1/2} = \frac{3}{2}\sqrt{x}$

b) Find the slope of the of the tangent line to the curve at  $(4,8)$ . Then find the equation of the tangent line to the curve at  $(4,8)$ .

$m = f'(4) = \frac{d}{dx}(x\sqrt{x})\Big|_{x=4} = \frac{3}{2}\sqrt{4} = 3$ . Therefore the equation of the tangent line is:

$y - 8 = 3(x - 4) \Leftrightarrow y = 3x - 4$

*Note: The bar to the right of  $dy/dx$  is read as “ $dy/dx$  evaluated when  $x=4$ ”.*

**NOTE:** Simplifying before trying to take a derivative is always a good idea.

We now present some fairly intuitive theorems which will help us differentiate a larger variety of functions.

**Theorem:** Suppose that  $c$  is a constant and  $f$  and  $g$  are differentiable functions. Then the following rules hold.

1. **The Constant Rule:**  $\frac{d}{dx}(c) = 0$

2. **The Constant Multiple Rule:**  $\frac{d}{dx}[c \cdot f(x)] = c \cdot \frac{d}{dx}[f(x)]$

3. **The Sum & Difference Rules:**  $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx}[f(x)] \pm \frac{d}{dx}[g(x)]$

*Proof (of 1):*

Suppose that  $f(x) = c$ . Then by the definition of the derivative:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{c - c}{h} = \lim_{h \rightarrow 0} 0 = 0$$

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*Proof (of 2):*

Suppose that  $g(x) = c \cdot f(x)$ . Then by the definition of the derivative:

$$g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{c \cdot f(x+h) - c \cdot f(x)}{h} = c \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = c \cdot f'(x)$$

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*Note: The proof of 3 (specifically the Sum Rule) will be an assigned problem for Homework Assignment 4.*

**Example 4:** Compute  $\frac{d}{dx}(x^7 - 22x^3 + 4x^2 - 6x + \sqrt{5})$

Using the above rules we have:

$$\begin{aligned} \frac{d}{dx}(x^7 - 22x^3 + 4x^2 - 6x + \sqrt{5}) &= \frac{d}{dx}(x^7) - \frac{d}{dx}(22x^3) + \frac{d}{dx}(4x^2) - \frac{d}{dx}(6x) + \frac{d}{dx}(\sqrt{5}) \\ &= \frac{d}{dx}(x^7) - 22 \frac{d}{dx}(x^3) + 4 \frac{d}{dx}(x^2) - 6 \frac{d}{dx}(x) + 0 \\ &= 7x^6 - 22 \cdot 3x^2 + 4 \cdot 2x - 6 \cdot 1 + 0 \\ &= 7x^6 - 66x^2 + 8x - 6. \end{aligned}$$

*Note: Normally, we will not show the application of each rule. In fact, as you get better at applying the rules of differentiation, you should be able to go right to the final answer in a problem like the example above.*

**Example 5:** Compute  $\frac{d}{dx} \left[ \frac{\sqrt{x}-2}{\sqrt{x}} \right]$

We begin by first simplifying the expression so that we can easily apply the given rules:

$$\frac{d}{dx} \left[ \frac{\sqrt{x}-2}{\sqrt{x}} \right] = \frac{d}{dx} \left[ \frac{\sqrt{x}}{\sqrt{x}} - \frac{2}{\sqrt{x}} \right] = \frac{d}{dx} \left[ 1 - 2x^{-\frac{1}{2}} \right] = 0 - 2 \cdot \left( \frac{-1}{2} \right) x^{-\frac{3}{2}} = x^{-\frac{3}{2}}.$$

**Exploration:**

We would like to compute the derivative of the exponential function  $f(x) = b^x$  where  $b$  is some positive real number constant. We note that the power rule does not apply here since the exponent is variable and not a constant. If we try to apply the definition of derivative we obtain:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{b^{x+h} - b^x}{h} = \lim_{h \rightarrow 0} \frac{b^x(b^h) - b^x}{h} = b^x \lim_{h \rightarrow 0} \frac{b^h - 1}{h}.$$

So, finding the derivative comes down to figuring out  $\lim_{h \rightarrow 0} \frac{b^h - 1}{h}$ , which unfortunately, is quite a difficult limit to compute.

Numerical explorations with the calculator seem to indicate that  $\lim_{h \rightarrow 0} \frac{2^h - 1}{h} \approx .693$  and  $\lim_{h \rightarrow 0} \frac{3^h - 1}{h} \approx 1.099$ . We

see that the derivative will be simplest when  $\lim_{h \rightarrow 0} \frac{b^h - 1}{h} = 1$ . So, we will focus on this situation for the remainder of the section.

*Note: The number,  $b$ , for which  $\lim_{h \rightarrow 0} \frac{b^h - 1}{h} = 1$  is one of the most famous irrational numbers of all time, and we now define it.*

**Definition (The number  $e$ ):** The number  $e$  is the real number such that  $\lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$ .

*Note: One may recall from Precalculus that  $e \approx 2.718281828459$*

*Note: The exploration and definition immediately imply the following theorem:*

**Theorem:**  $\frac{d}{dx}(e^x) = e^x$

**Example 6:** Suppose that  $f(x) = -2x + e^x$ .

- a) Find  $f'(x)$ .

Based upon what we have learned we calculate that  $f'(x) = -2 + e^x$ .

- b) At which point(s) on the curve determined by  $f$  is there a horizontal tangent line?

We know a horizontal line has a slope of 0, and the first derivative of  $f$  determines the slope of tangent lines to the curve. So, we solve:

$$f'(x) = 0 \Leftrightarrow -2 + e^x = 0 \Leftrightarrow e^x = 2 \Leftrightarrow x = \ln(2)$$

So the only point at which there is a horizontal tangent line is  $(\ln(2), -2\ln(2) + 2)$ .